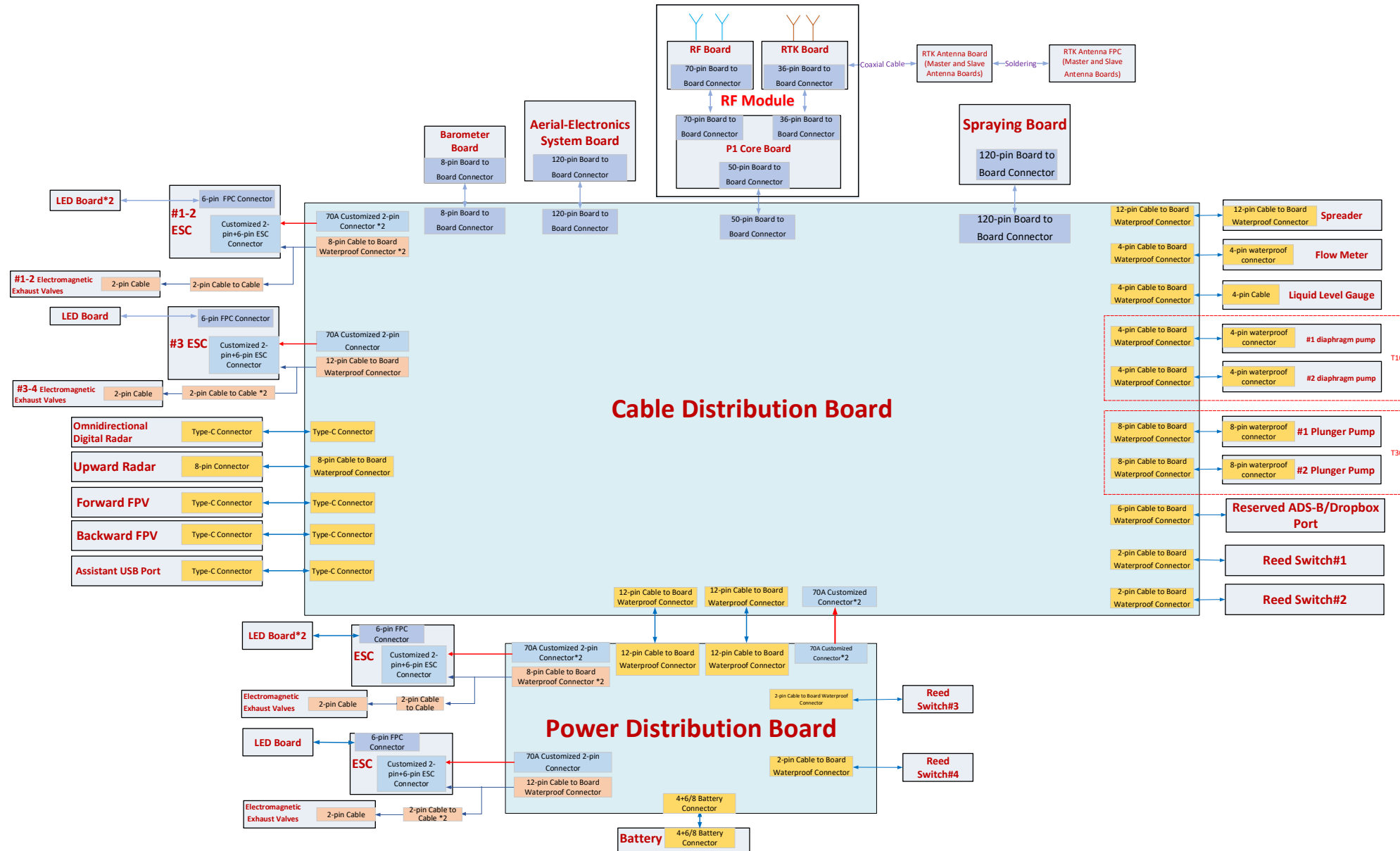


2022

T30 The Introduction of Damage Assessment Process

T30 Aircraft Principle Diagram--Review



T30 Aircraft Principle Diagram--Review

System	Module/Board	Function
Aerial-Electronics System	Aerial-Electronics System Module	The aerial-electronics board is the core module controlling the flight controller. It integrates the flight algorithm, image transmission, and flight controlling functions into one module. The compass is inside the module.
Propulsion System	Motor	The motors work with propellers to blow wind downwards and make the aircraft fly upwards. It also helps with spraying. The six motors are interchangeable.
	ESC Module	The ESC board converts the throttle information to the motor's rotation speed information, controls the motor rotation, and sends the motor information to the aerial-electronics board.
	LED	The LED board indicates the aircraft's current working condition. Users can know the aircraft's current working condition by the LED's color.
	Power Distribution Board	The power transfer module that distributes the power to the six motors, and is the transfer station for the battery and power supply.
	Cable Distribution Board	Connects the cables and modules and includes a barometer.
Spraying System	Barometer	Receives the current barometric pressure signal, sends the barometric pressure information to the aerial-electronics board, and detects the aircraft's current altitude information.
	Spraying System Module	The control system of the aircraft's core spraying function. It controls the pump and spraying volume.
	Liquid Level Gauge	For T30, it is a continuous liquid level gauge to measure the current liquid amount in the tank to notify the customer whether to add liquid. For T10, similar with T20.
	2-Channel Flow Meter	It is essential for high-precision spraying. It detects the pump's spraying speed. Therefore, the pump's spraying accuracy can be controlled.
	Delivery Pump	The water pump is the exerciser of the aircraft spraying system, which sprays pesticides out by rotating. For T30, the plunger pump is used, and diaphragm pump for T10.
Positioning System	Electromagnetic Exhaust Valves	Adjust the air in the water pipe, and can independently turn on or off any nozzle to spray.
	RF Module (Baseband Board, RF Board, RTK/GPS Board)	As a flight controller system to provide the aircraft position, direction, and speed of the aircraft to ensure the flight stability, and the efficiency and accuracy of the operation. It can track GPS L1/L2, Glonass F1/F2, BDS B1/B2, and Galileo E1/E5b frequency bands at the same time. It has positioning and orientation functions. The GPS module uses the main antenna of RTK module to receive GPS L1, and Glonass F1 frequency band signals.
Radar System	Active Antenna Board	Amplify and adjust the weak GPS signal so that the RTK/GPS board can get better satellite signals.
	Omnidirectional Digital Radar	A sensor for obstacle avoidance and navigation of aircraft. Responsible for investigating the external environment to prevent the aircraft from hurting people and crash.
Image Transmission system	Upward Radar	One of the sensors to detect the environment above the aircraft to prevent collisions with obstacles while ascending.
	Forward and Backward FPV Cameras	Converts optical signals into digital signals, and transmits them to customers through image transmission. The FPV camera contains LEDs to provide night lighting.
Battery System	RF Module (Baseband Board, RF Board, RTK/GPS Board)	Includes the baseband board and RF board, processing the image transmission signal and connecting the aircraft and remote controller.
	Battery	Batteries are the source of energy for flight and spraying. It is an energy storage device.



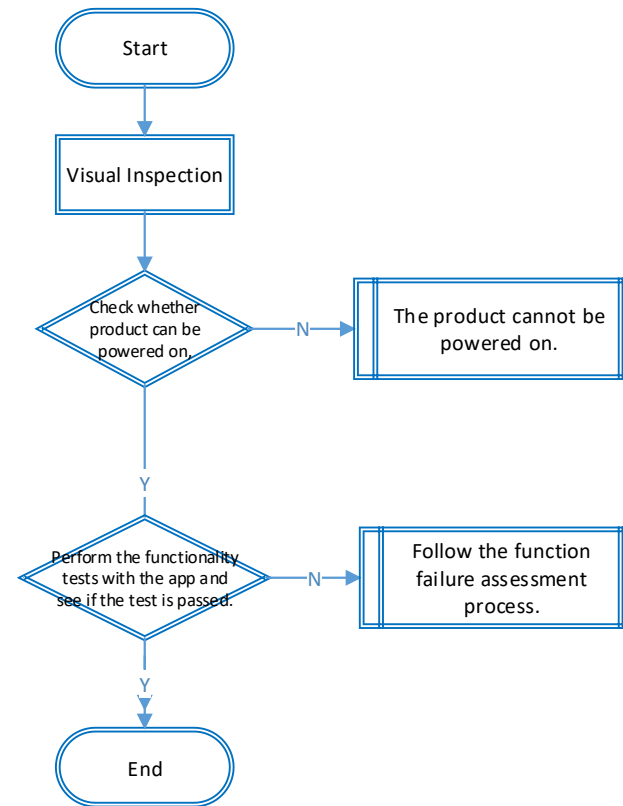
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Malfunction Assessment and Analysis

Malfunction Assessment and Analysis—Common Issues

Using HMS in the app for damage assessment.

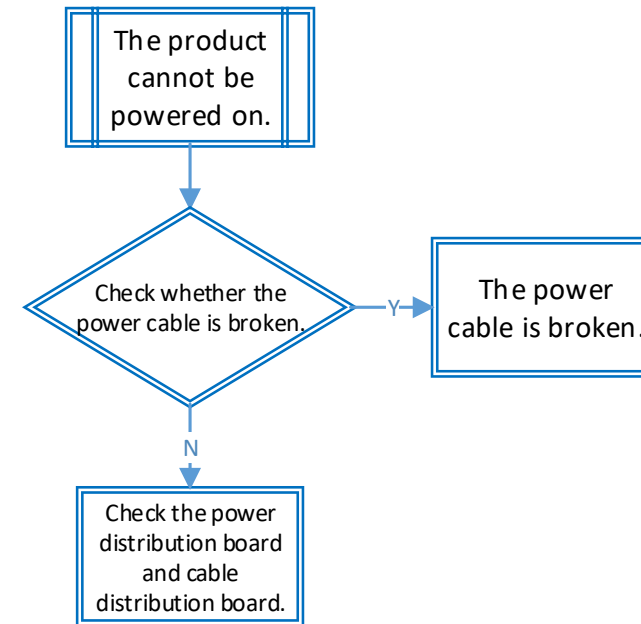
Check whether there is any cosmetic damage, whether the aircraft crashed, whether the boards are water damaged, and whether any component is incomplete.



Malfunction Assessment and Analysis—Common Issues

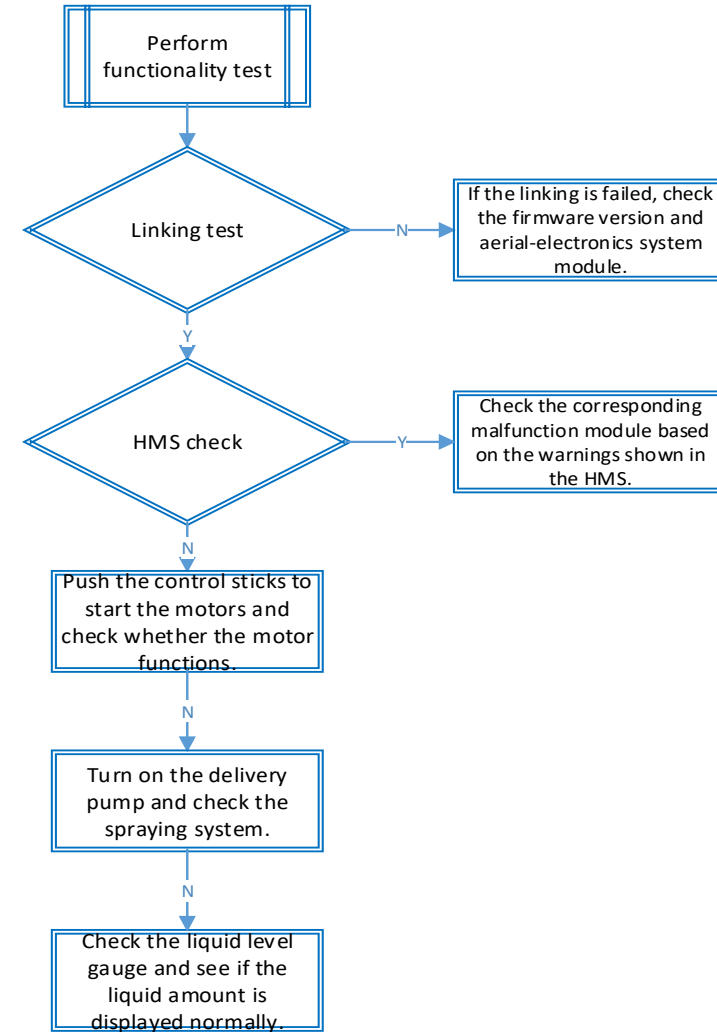
If the product cannot be powered on, check the following:

1. Main power
2. Power distribution board
3. Cable distribution board



Malfunction Assessment and Analysis—Common Issues

Link the aircraft with the remote controller and check whether each piece of hardware is normal via the app. If there are any warnings shown in the app or the module cannot be turned on because the module cannot be identified, determine that the corresponding module and cables are damaged.



List

Motor

Material Name	Material Number	T16	T20	T10	T30
10015 Motor (Thickness)	BC.AG.AA000137.02	✓	×	×	×
10015 Motor (Thin)	BC.PR.AA000090.01	×	✓	×	×
10010S Motor	BC.AG.AA000336.01	×	×	✓	×
10018 Motor	BC.AG.AA000335.01	×	×	×	✓
Silicone Rubber Pad	YC.XJ.QT000196.04	✓	✓	✓	✓
Teflon Gasket	YC.SJ.WS002416.03	✓	✓	✓	✓
Aluminum Alloy Propeller Holder CCW	YC.JG.QX000240.01	×	✓	✓	✓
Aluminum Alloy Propeller Holder CW	YC.JG.QX000238.03	×	✓	✓	✓
33-inch Propeller CCW	YC.SJ.GG000060.01	✓	✓	✓	×
33-inch Propeller CW	YC.SJ.GG000061.01	✓	✓	✓	×
38-inch Propeller CCW	YC.JG.ZS000813.01	×	×	×	✓
38-inch Propeller CW	YC.JG.ZS000812.01	×	×	×	✓

Aircraft

Module	T16	T20	T10	T30
Aerial-Electronics System Module	x	x	✓	✓
Spraying Module	x	x	✓	✓
RF Module	x	x	✓	✓
RTK Module	x	x	✓	✓
Power Distribution Board	x	x	x	x
Cable Distribution Board	None	None	x	x
Radar	x	x	✓	✓
Delivery Pump	✓	✓	x	x
Electromagnetic Exhaust Valves	None	x	x	x
FPV Module	x	x	✓	✓
Liquid Level Gauge	x	x	x	x
Upward Radar	None	None	x	x
ESC	x	x	✓	✓
Remote Controller	x	x	✓	✓



Thank You!